

ENVS 193DS Final

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Homework set up

```
#read in packages
library(tidyverse)
library(here)
library(janitor)
library(readxl)
library(ggplot2)
library(ggeffects)
library(flextable)
library(here)
library(MuMIn)
library(DHARMA)
library(rstatix)
#stored nest data as an object called nests
nests <- read_csv(here("data", "nest_data_final.csv"))
```

Problem 1. Research writing

a. Transparent statistical methods

In part 1, they used a logistic regression, which is a generalized linear model with a binomial error distribution. In part 2, they used a chi-square test, which is used to identify if there is a relationship between two categorical variables.

b. Suggestions for rewriting

Part 1.

The distance to water edge does seem to predict the probability of American Avocet microhabitat use.

Part 2.

There does seem to be a relationship between bird species (American Avocet, Forster's Tern, Black-necked Stilt) and habitat (managed pond, non-tidal marsh, salt pond, tidal marsh, or tidal mudflat).

c. Further analysis

One additional variable that could influence the probability of American Avocet microhabitat use is the type of behavior they were displaying, which is a categorical variable. For the American Avocets, the researchers put behavior into four main categories: feeding (searching, probing, and dunking bills into the water), roosting (standing, sitting, preening), breeding (nest-building, incubating, brooding chicks, alarm calling, breeding display, copulation, or predator distraction display), and other (walking, swimming, or flushing). This variable might influence American Avocet habitat use as different behaviors can change the duration of how long they remain in a particular habitat.

Problem 2. Data analysis

a. Clean and wrangle

```
#created new object from nests
nests_clean <- nests |>
#cleaned column names
clean_names() |>
#selected for tree height, ant_species, and case_control columns
select(case_control, ant_species, height_cm) |>
#reordered names in ant_species column
mutate(ant_species = as_factor(ant_species),
       ant_species = fct_relevel(ant_species,
                                "AB",
                                "RRB",
                                "BBR")) |>
#filtered ant_species column to only have three required species
filter(ant_species %in% c("AB", "RRB", "BBR")) |>
#created new column with full scientific name for each ant species
mutate(scientific_name = case_match(ant_species,
                                   "AB" ~ "Crematogaster sjostedti",
                                   "RRB" ~ "Crematogaster mimosae",
```

```

"BBR" ~ "Crematogaster nigriceps")) |>
#reordered full scientific name of ant species
mutate(scientific_name = factor(scientific_name,
                               levels = c("Crematogaster sjostedti",
                                           "Crematogaster mimosae",
                                           "Crematogaster nigriceps")))
#displayed structure of nests_clean data frame
str(nests_clean)

```

```

tibble [270 x 4] (S3: tbl_df/tbl/data.frame)
 $ case_control   : num [1:270] 1 0 0 0 0 1 0 0 0 1 ...
 $ ant_species    : Factor w/ 5 levels "AB","RRB","BBR",...: 2 2 2 2 1 2 3 2 1 3 ...
 $ height_cm      : num [1:270] 392 310 513 154 324 ...
 $ scientific_name: Factor w/ 3 levels "Crematogaster sjostedti",...: 2 2 2 2 1 2 3 2 1 3 ...

```

```

#displayed 10 random rows from the data frame but made it separate from all
↪ codes above by using separate pipe operator
nests_clean |> slice_sample(n = 10)

```

```

# A tibble: 10 x 4
  case_control ant_species height_cm scientific_name
      <dbl>   <fct>         <dbl> <fct>
1         0 AB             180 Crematogaster sjostedti
2         0 RRB            165 Crematogaster mimosae
3         1 RRB            396 Crematogaster mimosae
4         0 RRB            260. Crematogaster mimosae
5         0 RRB            104. Crematogaster mimosae
6         0 RRB            341 Crematogaster mimosae
7         0 RRB            175 Crematogaster mimosae
8         0 AB             130. Crematogaster sjostedti
9         0 AB             132. Crematogaster sjostedti
10        0 RRB            432. Crematogaster mimosae

```

b. Response variable

For the case_control column, the 1s mean that a tree does contain a nest and the 0s mean that a tree does not contain a nest (is an “available” tree for comparison).

c. Purpose of study

Tree height is a continuous (numeric) variable and could influence the probability of nest occurrence as if tree height were to increase, the probability of nest occurrence would increase because there would most likely be less predators higher up on trees (Results section, paragraph 2). Acacia ant species is a categorical (nominal variable) and could influence the probability of nest occurrence as if a particular ant species at a nesting site was more aggressive, the probability of nest occurrence could increase as it would reduce nest predation risks (Discussion section, paragraph 1).

d. Table of models

Model number	Tree height	Ant species	Model description
0	X	X	All predictors (no interaction term, +)
1	X	X	All predictors (interaction term, *)

e. Run the models

```
#model 0: all predictors, no interaction term
nest_model0 <- glm(case_control ~ height_cm + scientific_name,
  data = nests_clean,
  #response variable is binary so test should be logistic regression
  family = binomial)

#model 1: all predictors, interaction term
nest_model1 <- glm(case_control ~ height_cm * scientific_name,
  data = nests_clean,
  #response variable is binary so test should be logistic regression
  family = binomial)
```

f. Select the best model

```
#calculated AIC for each model
AICc(
  nest_model0,
  nest_model1) |>
#arranged output in order of AIC
```

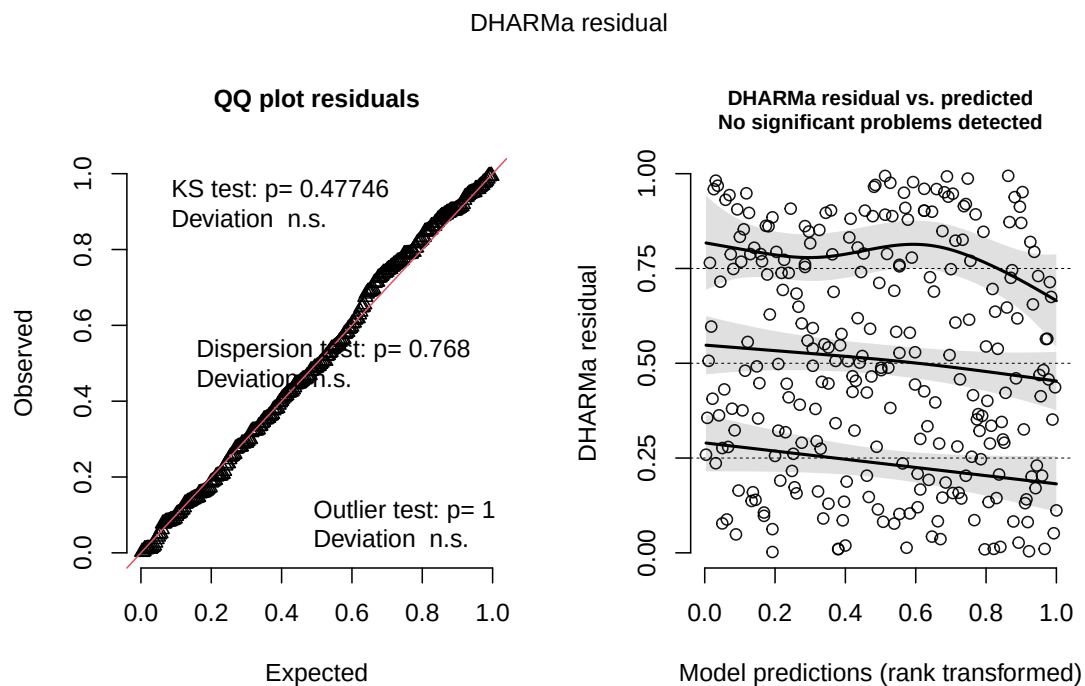
```
arrange(AICc)
```

	df	AICc
nest_model11	6	238.1236
nest_model10	4	258.8940

The best model as determined by Akaike's Information Criterion (AIC) was the model where tree height (cm) and acacia ant species were interacting terms for the probability of nest occurrence.

g. Check the diagnostics

```
#checked diagnostics for model 1 using DHARMA residuals  
plot(simulateResiduals(nest_model1))
```



Most of the points on the QQ plot follow the reference line, so we can assume the data is roughly normally distributed. We can only believe the observations collected were independent

and the response and we already know the response variable will be distributed following a binomial distribution (0 or 1). For the plot on the right, since it says there are no significant problems detected, we can also assume there is a linear relationship between transformed expected response (nest occurrence probability) in terms of link function and explanatory variables (tree height and ant species). Therefore, all of the assumptions for a generalized linear model (GLM) are met.

h. Visualize the model predictions

```
#created new object called preds
preds <- ggpredict(nest_model1,
#predictors
                    terms = c("height_cm[all]",
                              "scientific_name"))
#added scientific_name column for visualization
preds$scientific_name <- preds$group

#base layer: ggplot
ggplot(data = nests_clean,
       mapping = aes(x = height_cm,
                     y = case_control,
                     color = as.factor(scientific_name))) +
#relabelled x- and y-axes
  labs(x = "Tree Height (cm)",
       y = "Nest Occurrence Probability",
#colored by ant species
       color = "scientific_name") +
#changed colors for each ant species plot
  scale_color_manual(values = c("Crematogaster sjostedti" = "darkred",
                                "Crematogaster mimosae" = "navyblue",
                                "Crematogaster nigriceps" = "darkgoldenrod"))
  ↵ +
  scale_fill_manual(values = c("Crematogaster sjostedti" = "orangered",
                                "Crematogaster mimosae" = "blue",
                                "Crematogaster nigriceps" = "yellow4")) +
#used a different theme from default
  theme_minimal() +
#removed background
  theme(panel.background = element_blank(),
#removed major gridlines
        panel.grid.major = element_blank(),
```

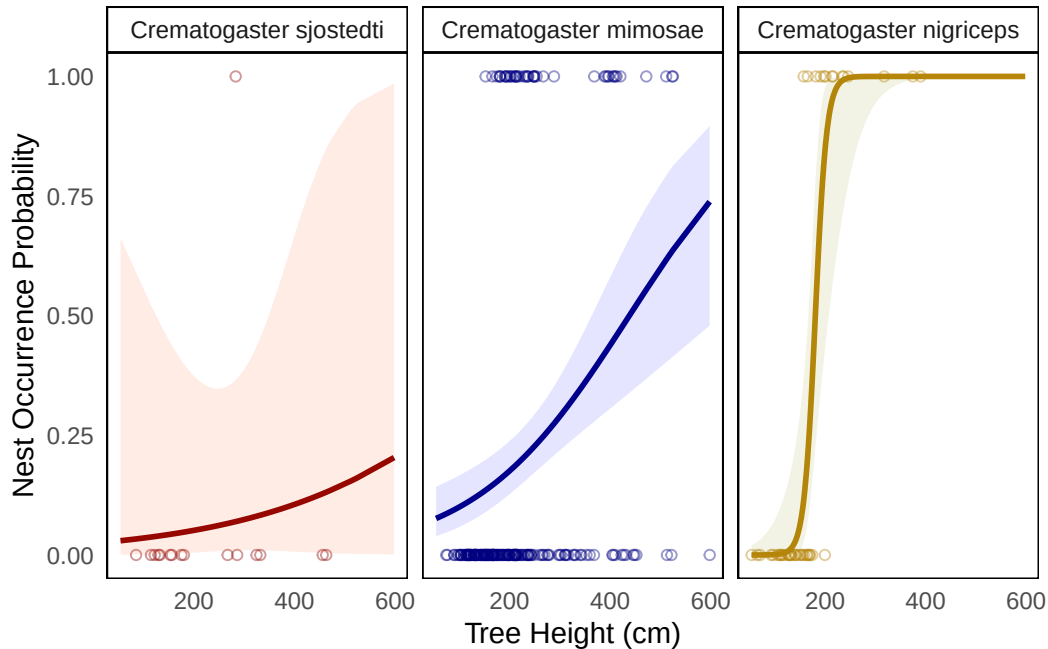
```

#removed minor gridlines
  panel.grid.minor = element_blank(),
#removed legend
  legend.position = "none",
#added borders to each panel
  panel.border = element_rect(color = "black",
                              fill = NA,
                              linewidth = 0.5),
#added borders to each ant species name title
  strip.background = element_rect(color = "black",
                                   fill = "white",
                                   linewidth = 0.5)) +

#first layer: geom point
  geom_point(mapping = aes(x = height_cm, #x-axis: tree height
                          y = case_control, #y-axis: nest occurrence
                          color = scientific_name), #set to selected colors

#changed shape
    shape = 21,
#made slightly transparent
    alpha = 0.4,
#ignored global aes for colors
    inherit.aes = FALSE) +
#second layer: geom line
  geom_line(data = preds, #data from model 1 prediction
            mapping = aes(x = x, #x-axis: tree height
                          y = predicted, #y-axis: nest probability
                          color = group), #colored by ant species
            linewidth = 1) +
#third layer: CI ribbon
  geom_ribbon(data = preds, #data from model 1 prediction
             mapping = aes(x = x, #x-axis: tree height
                           ymin = conf.low, #set low value
                           ymax = conf.high, #set high value
                           fill = group), #filled by ant species
             inherit.aes = FALSE, #ignored global aes
             alpha = 0.1) + #made transparent
#made new panel for each ant species
  facet_wrap(~ scientific_name)

```



i. Write a caption for your figure

Figure 1. Probability of bird nest occurrence varies based on the acacia ant species and height of the tree. Each panel represents the data for each of the three acacia ant species, which are differentiated by colors (red: *Crematogaster sjostedti*, blue: *Crematogaster mimosae*, gold: *Crematogaster nigriceps*). Points represent observations of nest occurrence based on the height of the tree (cm) for each acacia ant species. Lines in each panel represent the model predictions (predicted probability) for nest occurrence based on the height of the tree (cm) and ribbons represent the 95% confidence interval for each species. Data originally from: Lujan, E., Nielsen, R., Short, Z., Wicks, S., Nderitu Watetu, W., Malingati Khasoha, L., Palmer, T. M., Goheen, J. R., & Alston, J. M. (2023). Data, code, and metadata for: Symbiotic acacia ants drive nesting behavior by birds in an African savanna [Data set]. Zenodo. <https://doi.org/10.5281/zenodo.8373322>.

j. Calculate model predictions

```
#calculating nest occurrence from model1 for each ant species
ggpredict(nest_model1,
#predictors but set height to 600cm
  terms = c("height_cm[600]"),
```

```
"scientific_name"))
```

```
# Predicted probabilities of case_control
```

```
scientific_name: Crematogaster sjostedti
```

```
height_cm | Predicted |      95% CI  
-----  
        600 |         0.20 | 0.00, 0.99
```

```
scientific_name: Crematogaster mimosae
```

```
height_cm | Predicted |      95% CI  
-----  
        600 |         0.74 | 0.48, 0.90
```

```
scientific_name: Crematogaster nigriceps
```

```
height_cm | Predicted |      95% CI  
-----  
        600 |         1.00 | 1.00, 1.00
```

k. Interpret your results

If we were looking at the tallest tree, which is 600 cm, the predicted probability of nest site occupancy would be 0.2 (95% CI: [0.00, 0.99]) when *Crematogaster sjostedti* are present, 0.74 (95% CI: [0.48, 0.90]) when *Crematogaster mimosae* are present, and 1.00 (95% CI: [1.00, 1.00]) when *Crematogaster nigriceps* are present. In Figure 1, by looking at the lines from each panel, there seems to be a positive correlation for all three acacia ant species where as the tree height (cm) increases, the probability of a bird nest occurrence increases as well. However, this is more strongly displayed in trees that had either *Crematogaster mimosae* or *Crematogaster nigriceps* as their predicted values for nest occurrence probability for a tree at 600 cm were 0.74 and 1.00, respectively, which was much higher than a 600 cm tree that had *Crematogaster sjostedti* as its nest occurrence probability was 0.2. Therefore, acacia trees with either *Crematogaster mimosae* or *Crematogaster nigriceps* seem to be preferred by birds for nesting sites as those two ant species are a lot more aggressive defenders, reducing nest predation rates and thereby increase bird fitness in comparison to the less aggressive *Crematogaster sjostedti*. Furthermore, trees with *Crematogaster nigriceps* in particular usually have denser canopies because of the species' role in pruning apical buds, so this denser leafage makes it more appealing to birds looking to make nests in much more covered canopies.

Problem 3. Affective and exploratory visualizations

- a. Comparing visualizations**
- b. Designing an analysis**
- c. Check any assumptions and run your analysis**
- d. Create a visualization**
- e. Write a caption**
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- g. Sharing your affective visualization**

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